

Topic 3 – SEMINAR 3

Topic: Estimating the Housing Supply Price Elasticity for UK Regions
Format: Problem Set

Task: Ideally, each sub-group will have a pre-meeting to discuss the problem set and possible answers. If a pre-meeting is not feasible, sub-group members shall send their thoughts and ideas to the two pre-selected presenters. The pre-selected presenters then prepare a short PowerPoint presentation for the seminar.

Note: Sub-groups A will answer question a), sub-groups B will answer question b), sub-groups C will answer question c), and finally, sub-groups D will answer question d). **The total length of each presentation shall not exceed 7 minutes!**

Description of Problem and Questions to be Answered/Discussed:

You just graduated from the LSE. You were lucky enough to get hired by one of the largest American real estate development companies, Conner and Co. Development Company. The company, which specialised in large-scale residential developments across the globe, due to clever management survived both the Global Financial Crisis and then the COVID-19 pandemic. The company is now engaging in a new expansion path. You work in the company's research division and are responsible for research related to the UK housing market.

The CEO of your company has top secret ("insider") information from NextGen, a recent start-up that has invented some of the most revolutionary new technologies of the 21st century. In particular, they are in the process of developing a new computer—the NucTop—that features a number of highly innovative new technologies including a bendable and virtually paper-thin screen as well as an extremely powerful Nuclei-Nano chip, which – NextGen's CEO claims – is about 1000 times faster than existing computer chips. The NucTop also features the latest AI-technology.



(The NucTop)

Extremely relevant to your new company and you, NextGen is planning to build two gigantic new facilities in the UK with the objective of 'conquering' the British and European markets. The new

facilities will serve the entire European market (the CEO is not concerned about BREXIT imposed or other trade barriers for this type of product). Both facilities will create thousands of largely high-skilled jobs. The first facility – a **research facility** – is planned **at the fringe of Oxford**. The second facility – a **production facility** – is planned **at the fringe of Newcastle**. The average skill level of the employees for the later facility is considerably lower.

The CEO of your company is quite convinced that the two new facilities will create strong additional demand for housing in the two markets in question (and the good thing for your company is that *so far nobody else knows about these plans*). The trouble is that your (American) CEO knows next to nothing about the particularities of the local UK housing markets. The CEO is particularly worried about the prospect of elastic housing supply in the two markets (i.e., a more or less horizontal long-run supply curve) that might erode the potential profits of the possible development projects.

The CEO asks four working sub-groups to prepare short presentations that answer the following **questions**:

(The CEO mentioned that **the presentation of each sub-group at the Board Meeting should ideally be around 4-5 minutes but, in any case, it shall not exceed 7 minutes**. She said that she was planning to stop presenters who exceeded the allotted time limit.)

SUB-GROUPS A1 and A2:

a) How elastic is the supply of new housing in the UK in an international comparison?

Try to answer this question based on your existing knowledge of the UK housing market and based on a brief review of the relevant literature. Your CEO expects you to briefly summarize the findings of the relevant studies (in case there are any). She asks you to **be critical in assessing the empirical approaches and findings** and to caveat at the Board meeting in case the findings from the relevant studies do not seem entirely trustworthy (i.e., be critical in assessing the research studies and their methodologies etc.).

PS: Your CEO, who did an LSE Summer School course ages ago, vaguely recalls that her then professor at the LSE was very critical of the then available studies on supply price elasticities. She suggests that you double-check whether the empirical strategies employed in the studies you find in the literature are rigorous/'state of the art', that is, you should check whether the studies employ an instrumental variable strategy, and if not, what methodology they employ.

SUB-GROUPS B1 and B2:

b) What are the local factors that likely determine the supply elasticity of local housing markets in the UK? Would you expect there to be large differences across markets in the UK? Explain your reasoning making use of (urban/real estate) economic theory and the existing empirical literature.

PS: Your line manager—the Head of Research—having graduated from the LSE 15 years ago, recalls from an MSc REEF alumni letter a couple of years ago (prior to COVID) that Hilber and Vermeulen did some research on the impact of different types of local supply constraints on house prices in England. He suspects that paper might be useful and you should check it out.

SUB-GROUPS C1 and C2:

- c) The CEO heard from an old British friend (whom she met at the LSE Summer School ages ago) that an LSE professor, Christian Hilber, has another paper forthcoming in the Economic Journal (joint with Andreas Mense) that reports, somewhat hidden in the Appendix, long-run supply price elasticity estimates for English regions measured between 1981 and 2011. The paper is entitled ‘The Rise and Fall of the Price-to-Rent Ratio: Why Are Superstar Cities Different?’. Your CEO wonders whether the paper contains estimates that are useful to gauge the supply price elasticities of the Oxford and Newcastle areas. She also wonders what methodology the authors use. Ideally, your CEO would like to have ‘in house’ elasticity estimates for more recent years (post-2010) for the Oxford and Newcastle areas. However, time is of the essence. What challenges do you envisage estimating more contemporaneous supply price elasticities for Oxford and Newcastle? Would you advise your CEO to pursue in-house estimates for more recent years? Justify your reasoning.

SUB-GROUPS D1 and D2:

- d) The CEO is considering trying to secure large parcels of land close to each of the two planned facilities in order to carry out residential development projects. Would you recommend purchasing the land at both locations, only one location (which one?), or none? What other options could/should the CEO consider? Explain your reasoning (pros and cons of each option). What other particular factors, apart from the housing supply elasticity, should the CEO consider prior to making the decision? Your CEO expects you to provide concrete advice.